

Exercises on conservation laws

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1. Why is the decay $n \rightarrow p e$ forbidden even without postulating lepton-number conservation?

An example of Lepton Number violation process at LHC:

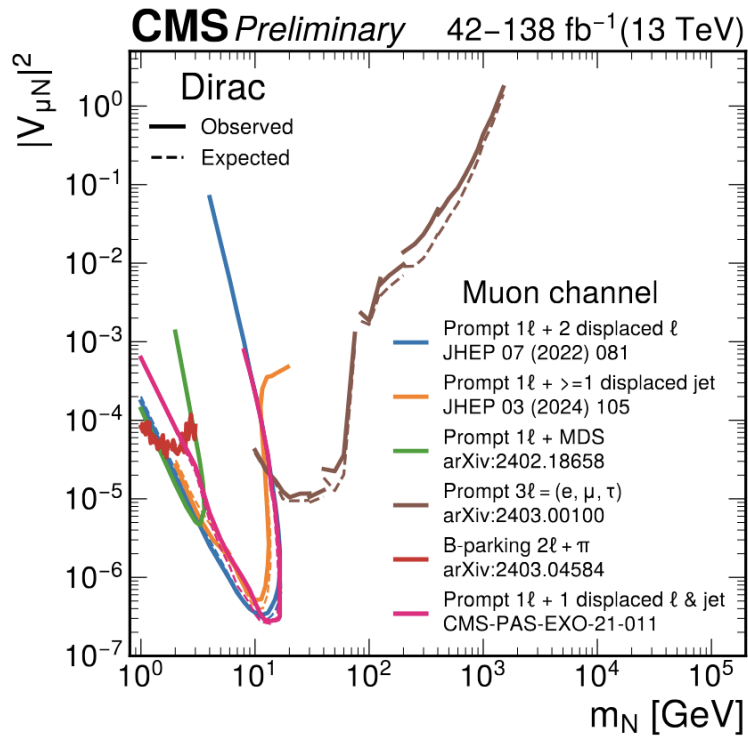


Figure 1: Taken from Physics Reports 1115 (2025) 570

2. The Δ^0 and Λ^0 particles decay into $\pi^- p$ with a mean lifetime of about 10^{-23} s and 10^{-10} s, respectively. Motivate this large difference. Explain how we do detect these particles and experimentally measure their lifetimes.

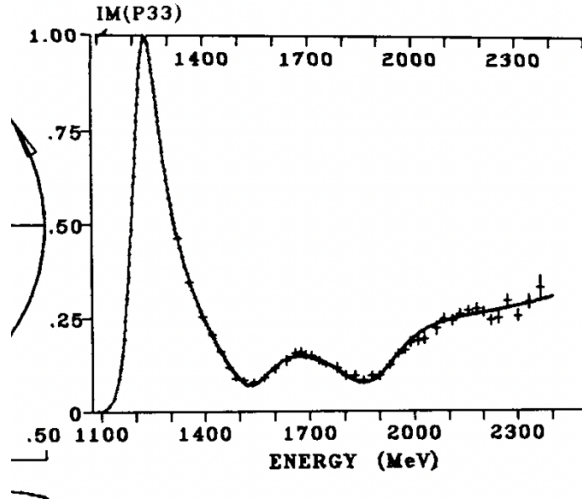


Figure 2: Taken from Cahn Goldhaber

3. Analyze the decay of the η , ρ , ω , ϕ mesons into two and three pions in terms of the P, C, I, and G symmetry.
4. Motivate why the ϕ meson decays into three pions only 15% of the times, while the remaining decays consist mostly of $\phi \rightarrow K K$.
5. Compare and comment the two following decays: $\phi \rightarrow \eta\gamma$ and $\phi \rightarrow \pi^0\gamma$
6. How do I look for new physics in B mesons decays?

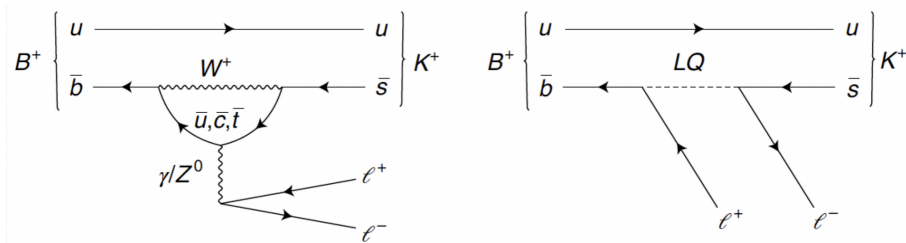


Figure 3: Taken from Symmetry 2023, 15(10), 1963

7. Why Σ^0 and $\Sigma^{+/-}$ have very different lifetimes?
8. Explain why $\Sigma^{*-} \rightarrow \Sigma^- \gamma$ is suppressed w.r.t. $\Sigma^{+*} \rightarrow \Sigma^+ \gamma$.